

Benjamin Zhao

Greater Austin |

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TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Go

Infrastructure & Cloud: OCI, Docker, Terraform, Linux

Testing & Automation: PyTest, CI/CD, Canary Testing

Data & Observability: Metric pipelines, Alerting, MQL, PostgreSQL

PROFESSIONAL EXPERIENCE

Oracle (OCI)

Software Engineer, OCI Compute Infrastructure | Austin, TX

July 2025 - Present

- Developed release-readiness automation (Release Coordinator Copilot) to synthesize validation status, failed-test evidence, rerun history, deployment context, and remediation guidance into decision-ready rollout summaries.
- Strengthened release-gating workflows coordinating 200+ automated test suites across service repositories, improving consistency and traceability of production-change reviews.
- Led automated security-patching rollout for compute infrastructure across 49 production hosts; paused rollout after correlating validation timeouts to a faulty software update using host evidence, rollback results, and failed deployment data.
- Built fleet-security reporting pipeline producing 22 validated monthly outputs; one cycle correlated 185 remediation tickets across 100,551 hypervisors and 478,520 virtual machines.
- Owned Oracle Linux 9 hypervisor-image validation, debugging rollout failures and separating product risk from test and environment noise before broader production deployment.
- Translated 248 federal security-hardening findings and FIPS readiness gaps into image-build changes, package validation, rollout evidence, and exception handling; reviewed 1,036+ packages and made 65 packages install-compatible.
- Expanded continuous validation to already-running virtual machines, detecting stability regressions and software-update failures missed by launch-only testing.

Software Engineer Intern Oracle Object Storage | Seattle, WA

June - Aug 2024

- Designed and deployed a metrics pipeline ingesting 50+ tenant-level object storage replication and copy-service signals for network and latency analysis.
- Authored 50+ metric-query monitors to isolate replication and networking issues, reducing fault localization time by ~40%.

Software Engineer Intern Oracle Virtual Machine Infrastructure | Austin, TX

May - Aug 2023

- Built a real-time server power-supply fault detection system, reducing mean detection time by ~60%, and enabling live repair services.
- Verified fault scenarios across 3+ hardware generations, supporting migration to non-terminating VM repair models.

Carleton College | Northfield, MN

Computer Science Course Staff and Lab Assistant

Sept 2022 - Sep 2024

- Mentored 50+ students in algorithms, computer systems, and computer architecture.
- Supported coursework on pipelining, cache design, and low-level performance reasoning.

PROJECTS

RoleFit Platform | *Personal Project*

May 2026

- Built a local Python system for ATS job-feed ingestion, deterministic role scoring, SQLite-backed tracking, and browser-based workflow automation.
- Implemented resume matching and tailoring pipelines with US/remote eligibility filtering, ATS deduplication, status workflows, and formatted DOCX export.

Computer Vision For Autonomous Driving | *Undergraduate Thesis*

Sep 2024 - Mar 2025

- Generated 1M+ multimodal simulation samples using CARLA with custom weather, vehicle density, and sensor configurations.
- Trained and evaluated HRNetV2 semantic segmentation models under domain shift conditions, recovering up to 95% mIoU and reducing fog-induced performance drops by ~35%.

Scheme Interpreter | *Personal Project*

Dec 2023

- Built a complete interpreter supporting primitives and continuations, strengthening parser, evaluator, runtime, and systems-level design fundamentals.

EDUCATION

University of Texas at Austin

Master of Science in Computer Science

Starting Aug 2026

Carleton College

Bachelor of Arts in Computer Science Concentration in Mathematics & Finance - (GPA 3.72)

Activities: Varsity Football, Club Rugby (**Forwards Captain & Treasurer**), Venture Capital Carleton (VP)

Graduated 2025

CERTIFICATIONS & AWARDS

Certifications: Oracle Cloud Infrastructure Certified Foundations Associate

Awards: 2023 CSC Academic All-District, 2022-2024 NCR Scholastic All-American, 2022-2024 MIAC Academic All-Conference